

OMER BIN HABIB

Software Engineer · Node.js, NestJS & TypeScript · PostgreSQL · Payments & reliability

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SUMMARY

Software engineer with six years of professional experience, specializing in backend systems with Node.js, TypeScript, PostgreSQL, and MySQL. Built financial and billing systems where correctness matters, including transactional loyalty-point transfers and Stripe Connect subscription billing. Experienced in designing reliable APIs, data flows, and business logic for reliable systems. Have also integrated LLM capabilities into applications, including model routing, structured outputs, vector search, and multi-agent workflows.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, NestJS, Express, REST API design, JWT authentication, input validation, Swagger / OpenAPI, queue-based processing (BullMQ, Redis), WebSockets (Socket.io), scheduled and background jobs, idempotency, retry and backoff, transactional consistency, bounded concurrency

Databases: PostgreSQL and MySQL — data modelling, normalisation, indexing, views, migrations, stored procedures, window functions, multi-tenant schemas with row-level security

Other server-side languages: PHP (Laravel, Lumen), Python (FastAPI)

Cloud & DevOps: AWS — Lambda, RDS, S3, EC2; Docker, GitHub Actions, nginx, Render, Vercel, Supabase, Sentry, Git

Design & practice: Object-oriented design and dependency injection through NestJS, design patterns, code review, automated testing (Jest / Vitest, Mocha, Playwright end-to-end), ESLint

Frontend: React, Next.js (App Router), Redux, TypeScript, Tailwind

AI & LLM: Claude, OpenAI and Gemini APIs; tiered model routing; schema-validated structured output (Zod); PII redaction before external calls; pgvector with hybrid semantic and keyword retrieval; LangGraph multi-agent workflows with Langfuse tracing; end-to-end testing with Playwright

Payments: Stripe Connect tiered billing

EXPERIENCE

Independent Consultant · Remote · Karachi, Pakistan

Jan 2026 – Present

Backend engineer for UK SaaS clients, working asynchronously with distributed clients

- **Spot — UK B2B workplace wellbeing SaaS.** TypeScript and NestJS backend. Stripe Connect tiered billing, timezone-aware processing, and data handling built for GDPR from the start with hard delete, anonymisation and export flows. Alongside it I built a Python FastAPI NLP service: PII redaction with spaCy named-entity recognition and language detection, outlier scoring with an Isolation Forest, and sentiment gating with sentence embeddings. 357 automated tests and 18 database migrations maintained across the life of the project.
- **AgencyView — reporting platform for a UK SaaS founder.** PostgreSQL-driven engine producing thirteen report types with output filtered by user role. Ingestion runs as a scheduled pipeline with bounded concurrency, retry policies and structured logging, so failures are diagnosable without manual inspection.
- **Reliability by construction:** idempotent writes so a repeated job never duplicates data, retry with backoff on external calls, and work moved onto queues rather than held in the request path.
- **Remote and asynchronous by default** — requirements, trade-offs and release scope agreed in writing with clients in different timezones, with no shared office hours.

Software Engineer · Blocship · Karachi, Pakistan

May 2023 – Dec 2025

Backend development across six production products, in a cross-functional team with frontend, product, design and QA

- **QR-code points and loyalty system for a vehicle-parts distributor.** NestJS backend handling point issuance, redemption and transfer between users — balance changes written transactionally so a failed transfer could not leave value in two places at once. I built the points reporting the distributor used to pay its buyers back according to the volume they had bought.

- **MaidMe — home services marketplace for providers in Norway.** The Laravel backend was already live and had no support for socket connections, so rather than rewrite it I added a dedicated NestJS socket server alongside it — bringing real-time booking management and in-app support chat to a running product.
- **Disgo — streaming rooms with a social platform.** Built the Express and MongoDB backend: room, feed and user APIs.
- **Groops — social platform for organisations.** NestJS alongside a Lumen service, covering group membership and engagement features — built on Clean Architecture: 16 NestJS modules, 73 migrations, 81 entities and 472 test suites.
- **Hubfiit and Apna Khata** — fitness and nutrition tracking; and an app for Pakistan's produce markets, where farmers sold fruit and vegetables to market traders — I built the transaction tracking between them. Laravel backends.
- **Across all six:** REST API design documented with Swagger, JWT authentication and input validation, MySQL and PostgreSQL data modelling and migrations, code review, and production support — reproducing reported faults, tracing them through logs and shipping fixes under live conditions.

Software Engineer · Sybrid (Pvt) Ltd · Karachi, Pakistan

Apr 2021 – May 2023

Enterprise web applications for Japanese and Pakistani clients

- **FITE — IT engineer hiring portal for Japanese companies.** Laravel, fully bilingual in English and Japanese, with role-based access across four user groups and security aligned to OWASP guidance.
- **MySQL stored procedures using window functions** to persist candidate profiles and calculate a profile score server-side, keeping scoring logic in one place rather than spread across request handlers.
- **Containerised a scheduled email service with Docker** — a reminder job for candidates who had begun but not completed an application, running behind nginx so the environment was reproducible rather than dependent on host configuration.
- **Multi-step profile form saved asynchronously per step**, so partial progress was never lost. Built with Bootstrap, jQuery and Ajax.

Associate Developer · TAFSOL Technologies · Karachi, Pakistan

Sep 2020 – Apr 2021

- **Digital sports agency platform** — Laravel, with payment gateway integration, an e-store, a training session video channel and an admin panel. Employee of the Month, October 2020.

SELECTED PROJECTS

ALLDost · Activity and nutrition tracking web application

Personal product

- **Next.js and React on a Supabase PostgreSQL backend** with row-level security, and an automated test suite maintained alongside it, including Playwright end-to-end tests.
- **Natural-language parsing of meals and workouts** into structured records using the Gemini API, with output validated against Zod schemas.

Multi-agent build harness · LangGraph and Langfuse

Personal project

- **Orchestrator, specialist and critic agents that build software**, traced in Langfuse. Used it to build resume-tailor (public on GitHub), a resume generator that only includes claims backed by a verified profile.

EDUCATION

Master's in Business Analytics (MSBA) · Karachi School of Business & Leadership

2025 – 2026

Final semester (In Progress)

BSc Computer Science · SZABIST · Karachi

Graduated 2020

CERTIFICATIONS

Introduction to Cloud Computing — IBM (2025) · Generative AI with Large Language Models — DeepLearning.AI with AWS · Microsoft Power BI · Building RESTful APIs in Laravel — LinkedIn Learning (2022)